

School of Computer Science and Engineering

COMP9511 Human Computer Interaction
Rental Rights Assistant

Group: Mon13-M13A-Real Human

Problem Statement and Product Description Statement

The Problem:

- Existing tenancy information platforms are static and complex
- Information is scattered across multiple pages and categories
- Users must identify their issue and navigate legal content themselves
- Legal language is difficult for non-expert users
- High cognitive load, especially for: First-time renters, International students, Users with low legal literacy

The solution:

- An interactive Rental Rights Assistant focused on user needs
- Allows users to describe their situation in plain language
- Provides a guided, scenario-based interaction flow
- Translates legal information into clear, understandable explanations
- Supports accessible and inclusive design, including: Simple language, Clear navigation, Reduced cognitive load

User Research

Research Method:

- Conducted an online questionnaire (Google Forms)
- Participants: students and general users
- Focus on: Understanding tenancy information, Experience with existing websites, Preferred support methods

Key Findings:

- Users struggled to locate relevant information
- Legal content was complex and difficult to understand
- Majority preferred guided, step-by-step assistance
- Users wanted clear, structured, and scannable information
- Phone-based support was inefficient and inaccessible

Evidence	Data / Insight	Analysis
Low satisfaction with existing platforms (Q11); difficulty finding useful info (Q9)	Average satisfaction **2.62 / 5**	Users struggled to locate relevant information, with satisfaction only around 2.6 out of 5.
Open-ended responses: too many rules, heavy legal language	Frequent complaints about complexity and legal jargon	Many users found legal content too complex, especially due to heavy legal language.
Guided assistant concept rated highly (Q14, Q15)	Guided flow ≈ **3.92 / 5**	Users preferred guided, step-by-step assistance, with a rating of around 3.9 out of 5.
Q10 & Q12: too much information, unclear navigation	Information overload and difficulty identifying relevant pages	Users reported that websites contain too much information, making it hard to find what is relevant.
Q8: users rely on friends and Google instead of official sources	Informal and scattered information-seeking behaviour	Many users rely on friend

Personas & Scenarios



Mei Lin (Age 22)

- Background:
- Chinese master's student
- First time renting overseas

- Issue:
- Can't understand legal English
- Heater broken 3 weeks
- Unsure of rights

- Needs:
- Multilingual support
- Plain language
- Clear next steps

- Scenarios:
Selects "Repairs"
→ answers simple questions
→ told landlord must fix urgently by law

Mandarin toggle available + pre-filled repair request template



Jake (Age 20)

- Background:
- Local uni student
- Just moved out of home

- Issue:
- Got rent increase notice
- Checked 5+ pages
- Gave up

- Needs:
- Short flow
- Quick answers

- Scenarios:
Selects "Rent increase"
→ answers lease type & duration

Can't raise rent mid fixed-term unless contract allows
→ response template

Margaret (Age 68)



- Background:
- Retired teacher
- Has presbyopia

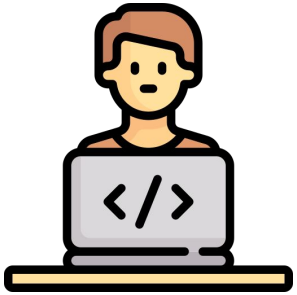
- Issue:
- Text too small
- No high-contrast mode
- relies on daughter

- Needs:
- Large font
- High contrast

- Scenarios:
Large text + high-contrast mode
→ taps through big labelled buttons

System explains notice period
+ right to dispute at NCAT (plain-language explanation)

Arjun (Age 30)



- Background:
- Indian developer
- Experienced renter
  but new to Australian system

- Issue:
- $800 bond withheld
- Couldn't navigate
  Bond Board / NCAT

- Needs:
- Jargon explained
- Step-by-step guidance
- Template letters

- Scenarios:
Selects "Bond dispute"
→ confirms condition reports exist

Step-by-step: negotiate → Bond Board
→ NCAT → template letter provided

Requirements

Functional Requirements:

- Guided question flow
- Plain-language information
- Clear actions and timelines
- Language support
- Accessibility settings
- Covers bond, repairs, rent, termination, and entry rights

Non-functional Requirements:

- Easy to learn and understand
- Accessible and inclusive interface
- Clear general-information disclaimer
- Professional and trustworthy visual design
- Fast task completion
- Privacy-preserving design

Initial Prototype and Testing

Figma Prototype

https://www.figma.com/proto/ZhH6bwIWNMPmG3W8AdvxkB/Original?t=LzhV8PqlSWEZZR3E-1

- Guided rental issue diagnosis flow
- Plain-language explanations with no legal jargon
- No audio dependencies, all feedback is visual
- Accessibility settings including font scaling, contrast, and translation
- Privacy-preserving interaction with no personal data storage

Usability Test

- Introduction and Consent
- Pre-Test Questionaires
- Think-Aloud Instruction
- 4 Tasks: Bond Dispute, Maintenance, Change text size, Switch language
- Post-Test Questionaires

Usability Test Results

Performance Results:

- 100% task completion rate across all tests
- 90% of tasks completed faster than expected benchmarks

Positive Feedback:

- Helpful, clear, simple, and trustworthy
- The overall flow easy to use
- The advice was useful and actionable
- Useful accessibility settings

Negetive Feedback:

- Hard to choose between similar situation options
- Language switch button was not prominent enough
- Prototype could be made more self-explanatory

Negative Issues Table

#	Severity	+/-	Issue	Description	Effects on Usability	Effects on User Experience	Priority
1	High	-	Consistency and Standards	The options 'Yes, in writing', 'Yes, by phone', and 'Contacted but no response' overlap semantically. A user who called but got no response fits multiple options, making selection confusing.	Efficiency, Effectiveness	Confusing, Frustrating	3
2	Med	-	Visibility	The three feature cards ('Understand Your Rights', 'Answer Simple Questions', 'Know What to Do Next') are redundant and misleading, describing internal process rather than distinct features.	Effectiveness, Learnability	Annoying, Boring	1
3	Med	-	Visibility	The webpage uses low contrast between text, cards, and the dark background. Important headings, labels, and buttons do not stand out clearly, making content harder to scan and interactive elements more difficult to identify.	Effectiveness, Learnability	Tiring, Frustrating	3
4	Low	-	Affordance	The header icons (question mark and gear) lack text labels. Users must guess what each button does, increasing cognitive load.	Learnability, Memorability	Confusing, Frustrating	1

Positive Issues Table

#	Severity	+/-	Issue	Description	Effects on Usability	Effects on User Experience	Priority
5	Med	+	Structure and Layout	The main issue selection page uses a clean card layout with generous white space, clear headings, concise descriptions, and example lists.	Effectiveness, Efficiency	Satisfying, Enjoyable	2
6	High	+	Constraints	Rental problems are categorized into four distinct types (Repairs & Maintenance, Rent Increase, Bond Dispute, Lease Ending) with icons and clear CTAs.	Efficiency, Learnability	Helpful, Motivating	3
7	High	+	Mapping	The Accessibility Settings panel provides text size options, contrast modes, and language selection (English, Chinese), accommodating diverse user needs.	Utility, Accessibility	Satisfying, Enjoyable	2
8	High	+	Feedback	Technical legal terms like 'NCAT' are accompanied by plain-English explanations, reducing jargon barriers for users without legal knowledge.	Learnability, Memorability	Helpful, Rewarding	2

Prototype Improvements Demo

Issue #1: Overlapping contact options.

- The contact method options overlap in meaning. For example, a user who called the landlord but got no reply may fit both “Yes, by phone” and “Contacted but no response,” making the selection confusing.

Issue #2: Misleading feature cards.

- The three homepage feature cards (“Understand Your Rights”, “Answer Simple Questions”, “Know What to Do Next”) describe internal process steps rather than clearly distinct features, which may make the interface harder to understand.

Issue #3: Low visual contrast.

- Text, cards, and buttons do not stand out clearly from the dark background, making content harder to scan and reducing readability.

Issue #4: Unclear header icons.

- The question mark and settings icons do not have text labels, so users must guess what each button does, increasing cognitive load.

https://www.figma.com/proto/LIha2ray8xulNwPJ9ZiyWE/New-System?node-id=0-1&t=wg4rr6jtuwixszTK-1

User-Centered Design Reflection

Positives:

- Evidence-based design decisions
- Iterative prototype improvement
- Strong focus on plain language, accessibility, and clear next steps

Negatives:

- Time-intensive research and testing process
- Limited participant sample
- Heavy workload across research, prototyping, testing, and revision

UCD helped us move beyond assumptions and design for real user needs. Through personas, scenarios, requirements, and usability testing, we identified issues such as overlapping options, low contrast, and unclear icons that may otherwise have been overlooked. This process led to a more accessible, trustworthy, and action-oriented Rental Rights Assistant.

Accessibility

Elderly / Visually Impaired Users:

- Adjustable text size helps users who find default text difficult to read.
- High-contrast mode and large, clearly labelled buttons reduce visual strain and make navigation easier.

Non-Native English Speakers:

- A multi-language option supports users who are more comfortable reading other languages than English.
- Plain-language explanations reduce dependence on complex legal terminology and make rental rights easier to understand.

Users Unfamiliar with the Australian Rental System:

- Terms such as NCAT, Fair Trading, and Rental Bond Board are explained in simple language on first use.
- Guided 4–6 step flows help users identify their issue and reach clear next-step advice without needing prior legal knowledge.

Conclusion

All Users:

• Plain-language explanations and 4–6 step guided flows reduce cognitive load and help users understand what to do next.

• Accessibility settings such as other language support, adjustable text size, and high-contrast mode make the prototype more inclusive for users with different language and reading needs.

• Large interactive elements, simple navigation, and clear next-step guidance support users with limited legal knowledge, visual needs, or unfamiliarity with the Australian rental system.

Results:

In conclusion, our system reduces cognitive load, improves accessibility, and provides clear next-step guidance.

Reflection:

Although every team member contributed many constructive design ideas, the workload was quite heavy and our timeframe was limited. As a result, we had to simplify or remove some planned features in order to complete the prototype on time.



Thank you !!!